

- MIRKO ENGLER, *Semantic determinacy and self-interpretations*.

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Drawing on ideas originating with [1], I motivate the philosophical standpoint that a syntactic interpretation of a theory T in a theory S determines the meaning of T relative to the meaning of S . In particular, an interpretation of T in itself determines its own meaning. This motivates the view that a theory may be regarded as *semantically determined* if there exists—up to a suitable notion of equivalence—exactly one self-interpretation. I then investigate the logical landscape generated by the various resulting notions of determinacy.

If a theory T admits, up to a (parametrically) in T definable homotopy (cf. [3, ch. 5.4]), only one self-interpretation, then T is called *a-categorical*. If the relevant homotopy is (parametrically) definable only within each model of T , then T is called *b-categorical*. In addition to regular interpretations, I also consider faithful interpretations, as well as interpretations arising as split monomorphisms or split epimorphisms, in the sense studied by [4].

I present a series of results relating this proposed terminology to established notions of semantic determinacy, such as categoricity and κ -categoricity, as well as to more recently introduced concepts, including those proposed by [2]. I also examine theories that are intuitively regarded as semantically determined. Among other results, I show that RCF and True Arithmetic are *a-categorical* with respect to normal interpretations, whereas PA and ZF are *a-categorical* when interpretations are taken to be split epimorphisms.

[1] RUDOLF CARNAP, *Logical Syntax of Language*, Kegan Paul, 1937.

[2] ALI ENAYAT AND MATEUSZ LELYK, *Categoricity-like properties in the first order realm*, *Journal for the Philosophy of Mathematics*, vol. 1 (2024), pp. 63–98.

[3] WILFRIED HODGES, *Model Theory*, Cambridge University Press, 1993.

[4] ALBERT VISSER, *Categories of theories and interpretations*, *Logic in Tehran. Lecture Notes in Logic*, (Ali Enayat, Iraj Kalantari and Mojtaba Moniri, editors), vol. 26, Cambridge University Press, 2006, pp. 284–341.